PHYS 4041/5041
Fall 2023
Homework 8
Due: In class, November 14, 2023

1. Consider a string that is clamped at $x = 0$ and $x = L$ that supports waves described by

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 u}{\partial t^2}, \tag{1}$$

where v is the wave speed. Solve this wave equation subject to the boundary conditions from clamping, $u(0,t) = u(L,t) = 0$, and the initial conditions

$$\begin{aligned} u(x,0) &= \begin{cases} \frac{hx}{a} & x < a \\ \frac{h(L-x)}{L-a} & x > a \end{cases} \\ \frac{\partial u}{\partial t}(x,0) &= 0, \end{aligned}$$

where h and a are constants.

2. An arbitrary function $f(\rho)$ defined for $0 \le \rho \le a$ can be expanded in Bessel functions:

$$f(\rho) = \sum_{n=1}^{\infty} A_{mn}\, J_m\left(x_{mn}\frac{\rho}{a}\right).$$

(a) Find the Parseval relation that expresses $\int_0^a |f(\rho)|^2\, \rho\, d\rho$ in terms of the A_{mn}.
(b) Determine the A_{mn} for $f(\rho) = \rho$ and $m = 1$.
(c) Use these results to show that

$$\sum_{n=1}^{\infty} \frac{1}{x_{1n}^2} = \frac{1}{8}.$$

3. A cylinder of radius a and length L has the curved surface held at a temperature T_0, with the ends held at constant temperature $T = 0$. Find the equilibrium temperature $\Psi(\rho, z)$ everywhere in the interior of the cylinder, i.e., solve $\nabla^2\Psi = 0$ subject to the boundary conditions. Note that the problem is azimuthally symmetric, so there is no ϕ dependence. Assume the cylinder is aligned with the z axis and that and that the ends held at zero temperature is located at $z = 0$ and $z = L$.

4. The Binomial Probability Distribution is given by

$$f_k = \binom{n}{k} p^k (1-p)^{n-k},$$

where n is an integer with $n \geq 0$, k is an integer with $0 \leq k \leq n$, p is a real number with $0 < p < 1$, and

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

The quantity f_k gives the probability of obtaining exactly k successes out of n trials, if the probability of success for each trial is p. One can show that

$$1 = \sum_{k=0}^{n} f_k \qquad \text{(normalization)}$$

$$np = \sum_{k=0}^{n} k f_k \qquad \text{(mean)}$$

$$np(1-p) = \sum_{k=0}^{n} (k-np)^2 f_k \qquad \text{(variance)}.$$

When $n \gg 1$, we can use Stirling's Formula to approximate the factorials provided that we avoid the extremes of the distribution, i.e., as long as $k \gg 1$ and $k \ll n$ as well. Show that this approximation leads (not unexpectedly) to the Gassian Probability Distribution:

$$f_k \to \frac{1}{\sqrt{2\pi np(1-p)}} \exp\left[-\frac{(k-np)^2}{2np(1-p)}\right].$$

5. Consider the steady-state heat equation, $\nabla^2\Psi = 0$, for the volume defined in cylindrical coordinates by $0 \le \rho < \infty$, $0 \le \phi < 2\pi$, and $z > 0$, i.e., half of all space. The boundary conditions are:

$$\Psi(\rho, z = 0) \equiv f(\rho) = \begin{cases} T_0 & \rho \le a \quad (T_0 \text{ is a constant}) \\ 0 & \rho > a \end{cases}$$

and $\Psi(\rho, z \to \infty) = 0$. Note that the problem is azimuthally symmetric and thus has no ϕ dependence. The boundary condition for $z \to \infty$ suggests that the z dependence of our solution should be $\exp(-kz)$. The other possibilities, $\exp(kz)$, $\sin(kz)$, or $\cos(kz)$, do not vanish at ∞ (I am assuming $k > 0$). Consequently, The full solution can be written as

$$\Psi(\rho, z) = \int_0^\infty A(k)\, \exp(-kz)\, J_0(k\rho)\, k dk,$$

a factor of k has been pulled out of $A(k)$ to make kdk, which puts the equation into the form of a Hankel transform. Note that k must be a continuous parameter in this case.

(a) Determine the Hankel transform of the boundary condition $f(\rho)$ defined by

$$B(k) = \int_0^\infty f(\rho)\, J_0(k\rho)\, \rho d\rho.$$

I expect you to evaluate the integral. It is an integral that you have encountered before. Note also that the inverse Hankel transform implies

$$f(\rho) = \int_0^\infty B(k)\, J_0(k\rho)\, k dk.$$

(b) Using your result for (a), determine the complete solution for $\Psi(\rho, z)$ that satisfies the Laplace equation and all boundary conditions.